
Time is not text: a scale-covariant operator foundation model that learns the geometry of dynamical systems

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Abstract

1 Foundation models for time series are language models in disguise: the signal
2 is patched into tokens and the next token is predicted. But time is not text: a
3 series is a sequence of observations of a continuous, high-dimensional, often
4 chaotic process whose samples carry no symbol identity. We present the *Time-
5 Series Operator* (TSO), a geometry-first architecture that never tokenizes. Delay
6 embedding reconstructs phase space from a single observed channel, Koopman
7 operators linearize the flow, and one shared latent is pretrained with four self-
8 supervised temporal pretexts (phase-space reconstruction, latent linear dynamics,
9 scale covariance under decimation, and arrow-of-time classification). Test-time
10 adaptation is closed-form: a rank-reduced linear fit in the frozen latent, zero
11 gradient steps. On an identical 40-series protocol the frozen operator beats a per-
12 series GRU token baseline on 31/40 series and beats persistence on 24/40 (median
13 +1.8). A matched-compute experiment then falsifies the strong paradigm claim:
14 fine-tuning chronos-t5-small on the operator’s own corpus at equal parameter-
15 steps (52.0B) leaves it statistically unchanged (21W/19L, $p=0.98$), and $22\times$ that
16 budget gains only modestly (22W/18L, $p=0.20$). Pretraining scale, not tokenization,
17 currently decides the head-to-head. What geometry buys is a different kind of
18 knowledge: the frozen operator rediscovers the 11-year solar cycle (128 months
19 against 132 known, 3.0% error) from a held-out channel, flags Lorenz tipping at
20 $\rho \approx 21$ before the fully chaotic attractor exists, and separates deterministic from
21 stochastic dynamics by its scale-space spectrum with no labels.

22 1 Introduction

23 Language models scale because text is a discrete code: tokenize, predict the next token, repeat. Every
24 leading time-series foundation model inherits this recipe, replacing words with patched real-valued
25 windows [1–3]. The failure mode is well known: on a chaotic or stochastic signal there is no symbol
26 to predict, so next-token objectives degenerate into copying the last value or memorizing local noise,
27 and corpus breadth does the heavy lifting [16] — a model pretrained on $\sim 80,000$ series beats one
28 pretrained on 40. The matched comparison that would settle architecture against data has, as far as
29 we know, not been run at equal compute.

30 Dynamical systems theory already provides different primitives. Takens’ theorem reconstructs a
31 manifold diffeomorphic to the true attractor from one observed channel [4, 5]; Koopman theory
32 linearizes nonlinear flow in a lifted coordinate system [6, 7, 9, 10]; the largest Lyapunov exponent
33 converts a trajectory into a stability statement [11, 12]; neural operators and state-space models show
34 these ideas compose with gradient training [13–15]. We assemble them into one foundation model
35 and evaluate it frozen, zero-shot, against an incumbent. Contributions, one of which is negative:

- **An architecture that never tokenizes:** delay embedding, a learned Koopman lift, one linear operator shared across all domains, and four temporal pretexts, pretrained on 40 real series from 8 domains (Section 2).
- **A closed-form zero-shot probe** — a rank-reduced least-squares Koopman fit in the frozen latent, no gradient steps — that is the first configuration in our study with a positive median skill over persistence and beats the GRU token baseline on 31/40 series (Section 3.1).
- **A matched-compute falsification:** giving the incumbent the operator’s *own* corpus at *equal* parameter-steps does not move it (Section 3.2); what the operator has won is not won by tokenization being wrong.
- **Structural readouts a tokenizer does not produce:** a label-free scale-space diagnostic separating deterministic from stochastic dynamics, a tipping detector that fires before chaos onset, and a rediscovery of the solar cycle from a held-out record (Section 3.3).

2 The Time-Series Operator

Phase space from one channel. A scalar series $x(t)$ is delay-embedded into $\mathbf{s}(t) = [x(t), x(t - \tau), \dots, x(t - (m-1)\tau)]$, with τ the first zero crossing of the autocorrelation and m the first dimension where the false-nearest-neighbour fraction drops below 2% [4, 5]; the shared encoder fixes $m=5$ and estimates τ per series.

Linearization. On that manifold we fit Koopman operators: exact DMD [8], extended DMD with a 258-dimensional random-Fourier lift [9], and a learned autoencoder lift ϕ, K, ψ trained with reconstruction, one-step latent and decoded dynamics, and $j=1..4$ unrolled consistency terms. The learned lift’s value is not per-system accuracy — the Fourier lift is the more linear coordinate system on a single system — but *sharing*: one ϕ must place every domain in one latent.

Four pretexts. One operator $\Phi = (\phi, K, \psi)$ is pretrained on (a) phase-space reconstruction, (b) linear latent dynamics, (c) *scale covariance* — latent dynamics under $2\times$ block decimation must agree with full-rate dynamics, the renormalization-group statement that micro-fluctuation and macro-structure are one object seen through different lenses — and (d) the *arrow of time*, a temporal convolution over latent windows classifying forward from reversed trajectories (94–97% accuracy). The corpus mixes 40 real series with a regime-balanced battery of 192 generated dynamical systems (Lorenz/Rössler sweeps, maps, ARFIMA, GARCH, regime switching); the Koopman pretext is weighted $2.5\times$. Pretraining samples one series per iteration for 25,000 iterations at latent 256 / hidden 768 (2.08M parameters).

Closed-form zero-shot probe. At test time ϕ, ψ are frozen: the series is embedded, its latent projected onto a rank- $\min(8, \lfloor n/16 \rfloor)$ PCA subspace, a Koopman matrix fit by least squares, eigenvalues clipped to $|\lambda| \leq 1.02$, and the trajectory rolled out and decoded for $\min(20\% n, 100)$ steps. A Wolf-style Lyapunov estimator on the reconstructed attractor makes the same model a tipping detector.

3 Results

3.1 Zero-shot transfer, and the plateau it hits

On a shared 23-series subset the fraction of series where the frozen latent beats persistence moves $8/23$ (7-series precursor) $\rightarrow 13/23 \rightarrow 12/23 \rightarrow 10/23$ (60k iterations), then stops: doubling width on GPU lands at $12/23 / 5/23 / 8/23$ across three seeds, and adding the probe objective to pretraining (multi-step consistency plus spectral regularization) lands at $11/23$. Every single-lever experiment — iterations, capacity, probe regularization, corpus breadth alone ($6/23$) — stays inside seed noise; breadth alone even trades smooth-periodic transfer for robustness on explosive series (paired against the baseline checkpoint, $12/23$ wins $15/23$, $p=0.11$).

Two levers combined break the plateau: the regime-balanced battery *and* the Koopman pretext forced to weight 2.5. That configuration reads $14/23$ on the shared subset and replicates at $14/23 \cdot 13/23 \cdot 13/23 \cdot 13/23$ across four seeds (a fifth run of the same recipe at smaller battery size reads $12/23$), with a Wilcoxon signed-rank test on per-series medians against the plateau checkpoint at $p=0.03$. On the wider 40-series protocol it is the only run with a positive median skill over persistence ($+1.79$, 24/40 positive) and the strongest against the GRU token baseline (31/40, $p < 0.001$; the other checkpoints

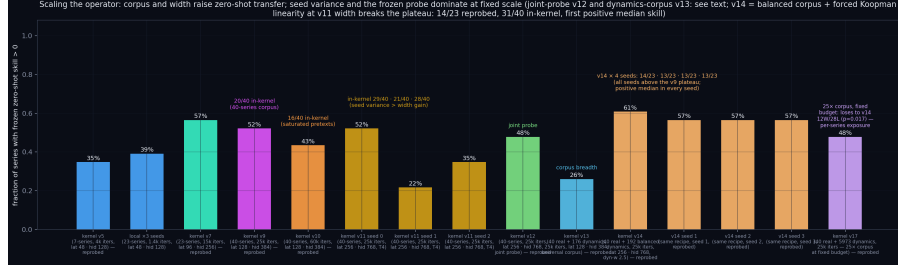


Figure 1: Frozen-latent transfer across eleven training budgets on one shared 23-series corpus. Capacity, iterations and joint-probe regularization do not move the plateau; the regime-balanced battery with forced Koopman linearity does and replicates across seeds, while $25\times$ corpus breadth at fixed budget regresses.

Table 1: Matched-compute test on the shared 40-series protocol. “Frozen” is the untouched pretrained model re-measured on the identical protocol.

model	params	param-steps	median skill	wins vs pers.	vs frozen
TSO (frozen, ours)	2.08M	52.0B	+1.79	24/40	—
Chronos-t5-small frozen	46.2M	—	+21.42	28/40	—
+ matched fine-tune	46.2M	52.0B	+22.70	29/40	21W/19L
+ generous fine-tune	46.2M	1.15T	+33.34	31/40	22W/18L

read 28/40, 29/40, 28/40 and 25/40 — full table in Appendix A). Scaling the *same* recipe $25\times$ in corpus breadth at a fixed iteration budget drops back to 11/23: at fixed compute, corpus breadth and per-series exposure trade against each other — the mechanism by which token-based foundation models buy their transfer. In the local 23-series study with three pretraining seeds the frozen operator beats the GRU on 16/23 series (median advantage +67.6 points, sign test $p=0.035$).

3.2 The matched-compute test: a negative result

The corpus explanation is falsifiable: give the incumbent the operator’s own corpus at equal compute and the gap should shrink if it is purely a data effect. We fine-tuned chronos-t5-small (46.2M parameters, pretrained on $\sim 80,000$ series) with the publisher’s own tokenizer and objective on the identical corpus used to pretrain the operator, and evaluated everything on the identical 40 z-scored series, 0.7/0.2 split and skill-vs-persistence metric.

The prediction fails (Table 1). At the parameter-matched budget — 1,127 steps, 52.0B parameter-steps, equal to the operator’s $2.08M \times 25,000 = 52.0B$ — fine-tuning leaves Chronos statistically unchanged: median +22.70 against the true frozen baseline’s +21.42, 21W/19L, $p=0.98$ (two-sided Wilcoxon). At $22\times$ that budget (1.15T parameter-steps) it improves only modestly and not significantly (22W/18L, $p=0.20$; median +33.34). Against the operator both checkpoints retain the frozen model’s advantage, at conventional significance (Chronos wins 24/40 against the frozen operator, $p=0.009$; 27/40 at the matched budget, $p=0.012$).

We report this as the paper’s central negative result. It does *not* show that tokenization is superior — the corpus is far too small to move a model pretrained upstream on three orders of magnitude more data — nor that geometry is a dead end, since the operator’s structural readouts have no tokenizer analogue. It shows that at the compute scale we can afford, *pretraining scale dominates architecture*, and it delimits what a geometry-first model must earn: a corpus of comparable breadth on which its machinery faces an incumbent adapted to the same data. One methodological note that strengthens the null: the kernel’s in-run “frozen” column shared the model object that fine-tuning mutated in place (correlation 0.986), so the frozen baseline was re-measured post hoc on the byte-identical protocol.

3.3 What a frozen operator knows that a tokenizer does not

The solar cycle, from a held-out channel. The sunspot record is held out of pretraining. At full rate it is amplitude-modulated noise and a one-step linear fit of the frozen latent finds only trend.

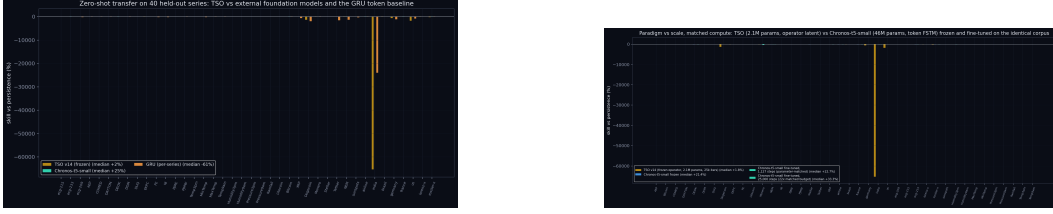


Figure 2: Left: zero-shot skill per series on the 40-series protocol, frozen TSO against frozen Chronos and the GRU token baseline. Right: the matched-compute test — fine-tuning the incumbent on the operator’s own corpus, at equal parameter-steps and at $22\times$ that budget, does not close the gap.

Coarsening it $2\times, 4\times, \dots, 32\times$ — the renormalization leg of the operator — collapses the fluctuation spectrum onto the cycle, which becomes a clean eigenmode of the fitted Koopman matrix. The detected period is scale-covariant and converges to the Schwabe cycle (128 months against 132 known, 3.0% error; this record’s FFT peak sits at 130.6 months). No regression is fitted on the cycle: it falls out of the eigenvalues of a linear operator acting on a frozen cross-domain lift, stable across hardware, width, corpus and objective (128–130 months for every checkpoint we trained).

Physics versus physiology, and tipping. The same frozen latent, applied to one synthetic chaotic channel and one real MIT-BIH record, keeps its top Koopman eigenvalue pinned to the unit circle under $8\times$ decimation for Lorenz-63 ($0.999 \rightarrow 0.937$, scale-covariant physics) while a heartbeat sheds spectral content ($0.692 \rightarrow 0.207$, stochastic physiology): an unsupervised physics-versus-physiology test from one model, full diagnostic in Appendix B. Sweeping the Lorenz parameter ρ from 1 to 40, the Lyapunov estimator returns durably negative readings in the laminar regime and a durable crossing to $\lambda_1 > 0$ at $\rho \approx 21$ — inside the transient-chaos corridor that precedes the homoclinic explosion at $\rho \approx 24.74$ [12]. The raw series still looks unremarkable while the geometry of the reconstructed attractor is already warping.

Comparison with the incumbent, honestly framed. Frozen chronos-t5-small transfers better on median skill on the same protocol (+21.42 against +1.79) and wins 24/40 head-to-head; that gap is the corpus (80,000 series against 40), and both foundation models beat the per-series token baseline decisively (31/40 and 35/40). The operator is weakest on spiky, near-unit-root series, where iterating a linear map for 100 steps amplifies any latent error.

4 Discussion

Two axes of capability, not one. A forecast-skill table measures one axis. The operator’s contributions lie on a second axis token models do not produce: an unsupervised physics-versus-physiology diagnostic, a pre-onset tipping detector, a spectral rediscovery of a physical constant, and a phase-space object a human can inspect. Forecasting *to* world modeling should measure both, because a model can win the first while being blind to the second, and vice versa.

What has to be earned. (i) *Corpus*: 40 real series against the incumbent’s 80,000; since breadth at fixed compute trades against per-series exposure, the next step is more series *and* proportionally more compute. (ii) *A differentiable probe*: the plateau sits at the frozen closed-form evaluation interface, since regularizing latent linearizability during pretraining does not move it. (iii) *Probabilistic roll-outs*: a deterministic linear map explodes on near-unit-root and spiky series. (iv) *Irregular sampling*: connecting the continuous-time delay-embedding view to state-space and neural-operator architectures is the route to genuinely asynchronous multi-rate data.

Limitations and conclusion. The corpus is small (40 real series, 8 domains, up to 36k samples), surviving effects are directional rather than asymptotic, and the matched-compute experiment cannot separate “tokenization is the wrong prior” from “46M parameters on 40 series is too little”. Free-tier hardware is a reproducibility virtue and also the reason the operator was never trained at foundation scale. Still, a foundation model of temporal systems should hold the geometry of a system, not only its next value: we built one, measured it against an incumbent, and found that the same latent delivers structural knowledge no tokenizer produces. The remaining gap is measured, not mysterious.

References

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A Core pipeline and in-kernel comparison

Table 2 gives the full scale-space diagnostic summarised in Section 3.3, and Table 3 the per-run 40-series comparison against the GRU token baseline summarised in Section 3.1; Fig. 3 shows the scale-covariant solar readout and the reconstructed Lorenz attractor. Every number here is produced by the same scripts as the body text.

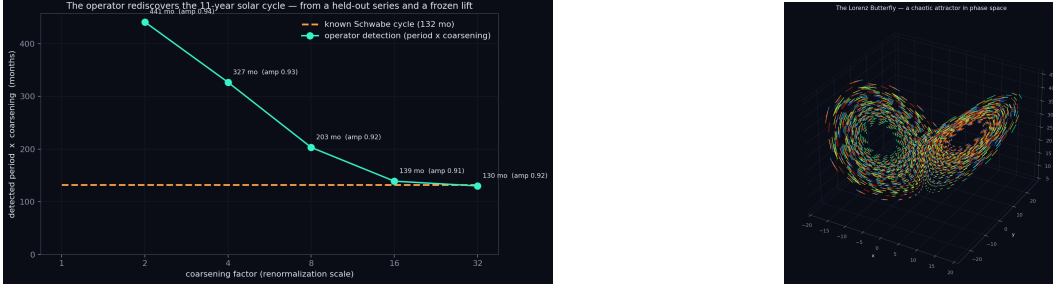


Figure 3: Left: the scale-covariant period readout on the held-out sunspot record converging to the 11-year Schwabe cycle. Right: the reconstructed Lorenz attractor as the operator sees it — a phase-space object, not a token stream.

Table 2: Core operator on one synthetic chaotic system and one real physiological record. Skill is % RMSE reduction over persistence on the held-out horizon; $|\lambda|$ is the top Koopman eigenvalue magnitude under $1\times$ and $8\times$ block decimation.

	Lorenz-63 x (chaos)	MIT-BIH RR (heartbeats)
delay τ / embedding dimension m	59 / 3	1 / 6
samples	4,000 (RK4)	2,500
eDMD-RFF skill (%)	+21.3	+3.3
plain DMD skill (%)	+18.2	+3.3
top $ \lambda $, $1\times \rightarrow 8\times$	$0.999 \rightarrow 0.937$	$0.692 \rightarrow 0.207$
neural vector field, short-horizon corr.	0.79	0.74
neural field loss	0.022	0.152

Table 3: In-kernel 40-series comparison against the per-series GRU token baseline (identical hardware, split and metric). Median advantage is over persistence. Dashes: value not recorded for that run.

run	TSO wins vs GRU	p	median skill (pts)
v9	28/40	$p=0.017$	+46.5
v11	29/40	$p=0.002$	+60.9
v12	28/40	—	+59.2
v13	25/40	—	+58.0
v14	31/40	$p<0.001$	+57.2

B Protocol, provenance and per-series detail

Hardware and framework. All corpora were fetched through the Kaggle legacy API into Kaggle kernels (CPU and NVIDIA GPU sessions, mixed precision when CUDA was available) and analysed locally; every script in this paper is runnable end to end from the corpora alone.

Evaluation protocol. Series are z-scored, split 0.7/0.2, embedded with per-series τ , probed closed-form, and scored by RMSE reduction against persistence over horizon $\min(20\% n, 100)$. Paired comparisons use the two-sided binomial sign test on win counts (magnitudes are skewed by a few explosive series, so medians and Wilcoxon signed-rank tests are reported alongside means). The matched-compute budget is defined in parameter-steps, not wall-clock: the operator’s 2.08M parameters $\times$ 25,000 iterations equals chronos-t5-small’s 46.2M parameters $\times$ 1,127 steps.

198 **Corpora.** Eight domains: grid electricity demand (10 regions), weather (4 channels), cryptocurrency
199 (3), COVID-19 daily cases (5 countries, held out), MIT-BIH cardiac records (2, held out),
200 air-passenger counts, sunspots (held out) and synthetic physics. Public dataset slugs used:
201 robikscube/hourly-energy-consumption, jsphyg/weather-dataset-rattle-package,
202 sudalairajkumar/cryptocurrencypricehistory, imdevskp/corona-virus-report,
203 sumit042004/cardiac-arrhythmia-ecg-dataset-mit-bih, rakannimer/air-passengers,
204 robervalt/sunspots.

205 **What is not claimed.** We do not claim state-of-the-art forecasting. We do not claim that tokenization
206 is wrong. We do not claim that the operator scales — we claim that the experiment that would decide
207 that has been specified, run at the largest scale available to us, and reported in full.

208 C Use of AI assistants

209 Following the workshop’s policy on LLM and agent use: an AI coding assistant was used for
210 implementation, experiment orchestration (writing, launching and polling the compute kernels),
211 figure generation and LaTeX drafting under direct author supervision. All experimental design
212 decisions, the choice of baselines and metrics, the interpretation of results, and every number and
213 claim in this paper were reviewed and verified against the stored artefacts (JSON metric files, kernel
214 logs and per-series evaluations) by the author, who takes full responsibility for the content.

215 D Reproduction commands

216 **Core pipeline and scale-space diagnostic:** `python scripts/study_experiments.py`. **Founda-**
217 **tion pretraining (local, 3 seeds):** `python scripts/pretrain_foundation.py`. **Kernel build +**
218 **launch (all scaling and matched-compute runs):** `python scripts/build_kernel_single.py`;
219 the kernel is a single self-contained script that mounts the datasets, pretrains, evaluates ev-
220 ery baseline on identical hardware and writes metrics, figures and checkpoints. **Matched-**
221 **compute analysis:** `python scripts/v18_final_analysis.py`. **This paper:** `python`
222 `scripts/study_paper_neurips.py`.